

Solution: Compute $\Delta^2 P(10)$

Given: A fourth-degree polynomial $P(x)$ and forward differences:

$$\Delta P(x) = P(x+1) - P(x)$$

with

$$\Delta^4 P(0) = 24, \quad \Delta^3 P(0) = 6, \quad \Delta^2 P(0) = 0$$

We are asked to compute $\Delta^2 P(10)$.

Step 1: Express $P(x)$ as a general polynomial

Let

$$P(x) = a_4 x^4 + a_3 x^3 + a_2 x^2 + a_1 x + a_0$$

Step 2: Relate differences to coefficients

For a 4th-degree polynomial, the forward differences at $x = 0$ are:

$$\Delta^4 P(0) = 24a_4 = 24 \implies a_4 = 1$$

$$\Delta^3 P(0) = 24a_4 + 6a_3 = 6 \implies 24 + 6a_3 = 6 \implies a_3 = -3$$

$$\Delta^2 P(0) = 12a_4 + 6a_3 + 2a_2 = 0 \implies 12 - 18 + 2a_2 = 0 \implies a_2 = 3$$

Thus, the polynomial (coefficients we need) is:

$$P(x) = x^4 - 3x^3 + 3x^2 + a_1 x + a_0$$

Note: a_1, a_0 are not needed to compute $\Delta^2 P(x)$.

Step 3: Compute $\Delta^2 P(10)$ using the forward difference formula

$$\Delta^2 P(x) = \Delta^2 P(0) + x \Delta^3 P(0) + \frac{x(x-1)}{2} \Delta^4 P(0)$$

Substitute $x = 10$:

$$\begin{aligned} \Delta^2 P(10) &= 0 + 10 \cdot 6 + \frac{10 \cdot 9}{2} \cdot 24 \\ &= 60 + 45 \cdot 24 \\ &= 60 + 1080 \\ &= 1140 \end{aligned}$$

Answer

$$\boxed{\Delta^2 P(10) = 1140}$$

Key Insight: Forward differences act like discrete derivatives. Higher differences contribute to lower ones as x increases. This method is faster and avoids constructing the full polynomial.